Label-Free, Dependency-Aware Channel Selection for Coupled Multi-Channel Sensor Data

Narek Meloyan[1,2,3], Narine Sarvazyan[1,2,3,4]
[1]Zaven P. and Sonia Akian College of Science and Engineering, American University of Armenia, Yerevan, Armenia
[2]Engineering Research Center, American University of Armenia, Yerevan, Armenia
[3]L.A. Orbeli Institute of Physiology NAS RA, Yerevan, Armenia
[4]George Washington University, Washington, DC, USA
Email: narek_meloyan@edu.aua.am, phynas@gwu.edu

***Abstract*—Dimensionality reduction is a central step in high-dimensional pipelines, from multi-channel sensor arrays and biomedical imaging to interpretable machine learning. Without labels this usually relies on methods such as PCA or t-SNE, which optimize label-free proxies (variance or neighborhood structure) rather than class discriminability, and can discard a weak discriminative signal dominated by redundant, high-variance channels. This paper presents a label-free, dependency-aware method that keeps a subset of the original channels rather than a projection, so the result stays interpretable. A group-structured autoencoder reconstructs all channels; each latent factor is perturbed, and the resulting per-channel reconstruction change gives a label-free relevance score that a relevance-redundancy rule turns into an ordered shortlist, changing only the encoder between modalities. Judged by a downstream classifier on the selected channels, the method matches a supervised mutual-information selector without labels, and beats PCA and variance ranking where the discriminative signal is not the loudest. On a controlled benchmark, where the informative channels are also the highest-variance ones, PCA recovers them too and the method matches the supervised selector, while variance and random selection lag far behind. On motor-imagery electroencephalography (EEG) the informative rhythm is weaker than the background, so PCA and variance fall to about random, while the method stays above all of them and matches the supervised selector using six of 22 electrodes. On biomedical hyperspectral imaging the same method keeps or improves accuracy while removing up to 95% of the bands. By accurately modeling channel dependencies without data labels, this architecture provides a powerful, interpretable alternative to traditional dimensionality reduction in complex sensor arrays.**

***Index Terms*—unsupervised feature selection, autoencoders, dependency-aware selection, interpretability, brain-computer interfaces, hyperspectral imaging.**

I. Introduction

Complex sensors produce many channels that overlap heavily and arrive in natural groups. For cost, power, bandwidth, and interpretability one often wants to keep a handful of the actual channels rather than a learned mixture of them, and the difficulty is deciding which, when no labels are available.

Unsupervised filters such as variance ranking or principal-component analysis (PCA) score each channel by its own marginal statistics. They are fast but do not model how channels depend on one another, and being variance-driven they fail when the informative signal is not the loudest channel. Supervised, dependency-aware selectors (mRMR, JMI) model conditional relevance but need labels. Few methods are at once label-free, dependency-aware, and discrete, returning real channels rather than a projection. The method proposed here occupies that niche; its central value is modeling channel dependence without labels. Where the discriminative channels are the loudest, marginal methods such as PCA recover them; where they are not, modeling how channels depend on one another separates the informative channels from the loud, redundant ones.

The presented approach is a label-free, dependency-aware selector that returns real channels, built from a group-structured autoencoder with a compressive bottleneck and perturbation-based attribution. A study of downstream classification accuracy across four modalities shows that its channels match a supervised selector without labels, and beat PCA and variance where the discriminative signal is not the loudest; a simple label-using check completes the account by marking, in advance, the redundant datasets on which selection of any kind offers little.

II. Related Work

Channel and feature selection methods fall into filter, wrapper, and embedded families. Variance ranking and the Laplacian score [4] are unsupervised but marginal. Dependency-aware selectors such as mRMR and JMI [2], [3] model conditional relevance but are supervised. Among unsupervised deep methods, the concrete autoencoder [1] learns a differentiable discrete selection layer, and in hyperspectral imaging attention networks such as BS-Net [5] rank spectral bands. In EEG, channel selection [6] is mostly supervised, and wearable human-activity recognition is also widely studied [7]. On redundant data, random selection is a strong baseline that many unsupervised selectors fail to beat [8]; the check proposed here makes that measurable.

III. Method

A. Problem and Group Structure

A multi-channel observation is organized into G groups (a positive integer). The grouping is the sensor's physical acquisition metadata, that is, one group per inertial unit, excitation wavelength, scalp region, or process subsystem. It is therefore known in advance and uses no class labels,

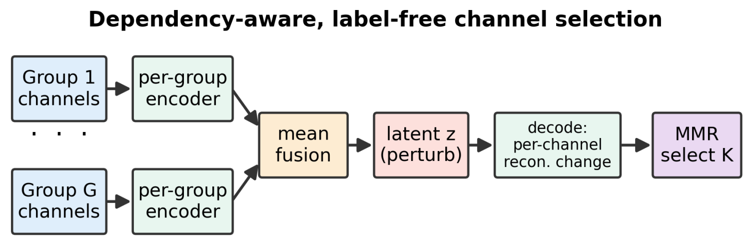

Fig. 1. The three-stage pipeline. A group-structured autoencoder is trained by reconstruction; latent perturbations give a per-channel relevance (the reconstruction change $I_{g,c}$ in Eq. (2)); and a relevance-redundancy rule selects an ordered subset. The engine is the same across domains; only the encoder and decoder change to match the regular axis of the data.

and the method also runs with $G{=}1$ when no grouping is available. Group g contributes C_g channels over a regular axis (time or space) of length T, so a windowed sample is the matrix $X_g \in \mathbb{R}^{T\times C_g}$ and the full sample the tuple $X = (X_1, \ldots, X_G)$, of which there are N in the dataset. The goal is an ordered subset $S \subseteq \{(g,c)\}$ of size $|S| = K$ (the channel budget, a positive integer) that is informative and non-redundant, using no labels.

B. Compressive Group Autoencoder

Each group is encoded independently, the per-group features are fused by averaging, and the code is decoded per group:

$$z \coloneqq \frac{1}{G}\sum_{g=1}^{G} E_g(X_g), \qquad \hat{X}_g \coloneqq D_g(z), \tag{1}$$

where E_g, D_g are the per-group encoder and decoder and $z \in \mathbb{R}^d$ is a latent vector of size d. The encoder is the only modality-specific part: 1-D temporal convolutions for signals and 2-D spatial convolutions for images. The encoder pools the regular axis to a per-window bottleneck, so z has no time or space extent. This keeps the code from absorbing per-channel noise, which is necessary for the attribution below to remain a reliable importance signal on non-image data. Training minimizes reconstruction error only, $\mathcal{L} \coloneqq \sum_{g=1}^{G} \|X_g - D_g(z)\|_2^2$, with no labels.

C. Perturbation Attribution and Selection

The d latent coordinates are ranked by their variance across the N samples; the top m are kept. For a retained coordinate $j \in \{1,\ldots,m\}$ and a magnitude δ from a finite set $\mathcal{D}$, the code is shifted $z^{(j,\delta)} \coloneqq z + \delta\, e_j$ (with e_j the j-th canonical basis vector), and the scalar relevance of channel c in group g is the accumulated reconstruction change:

$$I_{g,c} \coloneqq \sum_{j=1}^{m}\sum_{\delta\in\mathcal{D}} w_j \,\frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T}\Big|\big[D_g(z^{(j,\delta)}) - D_g(z)\big]_{n,t,c}\Big|, \tag{2}$$

averaged over the N samples and T axis positions, with scalar weight w_j scoring coordinate j by its variance rank. A channel whose reconstruction is sensitive to the latent factors is one the joint representation depends upon. Relevances are normalized within group, $\tilde{I}_{g,c} \coloneqq I_{g,c}/\max_{1\le c'\le C_g} I_{g,c'}$, and channels are chosen greedily by a relevance-redundancy rule, i.e. maximal marginal relevance (Fig. 1):

$$c^{\star} = \arg\max_{(g,c)\notin S}\Big[\tilde{I}_{g,c} - \lambda \max_{s\in S}\operatorname{sim}\big((g,c),s\big)\Big], \tag{3}$$

with sim the cosine similarity between channel profiles and scalar $\lambda \in [0,1]$ the redundancy penalty; the output is ordered and can be truncated to any budget K.

IV. Experimental Setup

Selection is always unsupervised; labels are used only by a downstream classifier to score the selected-K channels (accuracy or macro-F1) against variance, PCA-loading, a supervised mutual-information (MI) selector, and random selection.

Controlled benchmark. Grouped time-series with four independent latent factors, each in a cluster of five correlated channels of decreasing amplitude, plus 44 pure-noise channels (64 channels, 16 classes); covering all four factors is required to classify. A k-nearest-neighbor classifier scores the subset.

Motor-imagery EEG. BCI Competition IV-2a [9]: 22 electrodes in four scalp-region groups, four classes, nine subjects, band-passed to 8 to 30 Hz. Because motor-imagery patterns are subject-specific, the standard within-subject protocol is used (train session 1, test session 2) with a common-spatial-patterns (CSP) and LDA classifier on the selected electrodes.

Biomedical imaging. Multi-excitation fluorescence cubes (lichen tissue, 192 bands, four classes; collagen sponges, 158 bands, three classes) with a spatial encoder, verified by per-pixel accuracy.

V. Results and Discussion

A. Selected-Channel Accuracy in Two Regimes

Table I reports the downstream classifier's score on the channels each method selects. The label-free method stays close to the supervised MI selector in both regimes (within 0.05 accuracy on synthetic and 0.01 macro-F1 on EEG) without labels, and above variance and random. The marginal methods differ between the two regimes. On the synthetic benchmark the informative channels lie in the high-variance subspace, so PCA recovers them (0.95, close to supervised); variance ranking, however, covers only half the factors and stays near random (0.24). On EEG the informative mu/beta rhythm is not the loudest signal, since alpha and ocular

TABLE I
DOWNSTREAM CLASSIFIER SCORE ON THE SELECTED CHANNELS; NO LABELS ARE USED FOR SELECTION.

Selector	Synthetic acc., K=8/64	EEG F1, K=6/22
All channels (full set)	0.87	0.58
Supervised MI	0.99	0.53
Proposed (label-free)	0.94	0.52
PCA	0.95	0.48
Variance	0.24	0.47
Random-K	0.25	0.49

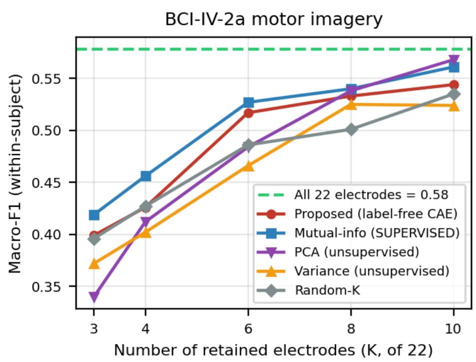

Fig. 2. Motor-imagery EEG (BCI-IV-2a, within-subject, CSP and LDA). At six of 22 electrodes the label-free method (red) is above PCA (purple), variance (orange), and random (grey), and close to the supervised selector (blue). PCA is no better than random at small budgets and catches up only at larger ones.

activity are stronger, so PCA and variance do no better than random (0.48 and 0.47 against 0.49) and select loud but uninformative electrodes, while the proposed method recovers the informative ones. On the synthetic benchmark, eight of 64 channels beat the full noisy set (0.94 against 0.87), so selection also denoises, though the bottleneck occasionally under-encodes a factor (mean 0.70 ± 0.29 over draws).

B. Motor-Imagery EEG

Figure 2 shows the result across budgets. At six of 22 electrodes the label-free method reaches 0.517 macro-F1 (about 89% of the full 0.578), above PCA, variance, and random and close to the supervised MI selector. PCA is no better than random here: it selects the loud alpha and ocular channels and misses the sensorimotor cortex, which the reconstruction-based relevance finds. PCA catches up only at larger budgets, and the method is at or above random for eight of the nine subjects.

C. Biomedical Imaging and the Boundary

On multi-excitation fluorescence cubes the same method (with a spatial encoder) gives large reductions that act as denoising: nine of 192 bands reach 89.4% accuracy, a 95% reduction and above the 88.2% full-band value, and 30 of 158 bands raise accuracy from 79.8% to 85.6%. These wins share a structure: the informative channels are neither redundant copies nor the loudest, and they are a small fraction of the input (four of 64 synthetic factors, nine of 192 Lichens bands), so pruning the rest is where label-free selection gains most. Because that advantage grows with the uninformative fraction of the input, these modest benchmarks are a conservative test for a method aimed at much larger channel arrays. Selection offers little only at the opposite extreme, where channels are so redundant that every subset carries the same information, as on wearable human-activity recognition (PAMAP2, Opportunity) and adjacent-band remote-sensing imagery (Indian Pines): there no unsupervised selector, the proposed one or PCA, separates from a random subset, and a supervised selector is needed [8]. A label-using check (best K versus random K under a supervised score) flags such cases before any modeling.

VI. CONCLUSION

This paper presented a label-free, dependency-aware channel selector: a compressive group autoencoder with perturbation attribution and relevance-redundancy selection. Measured by downstream classification accuracy, its channels match a supervised selector without labels and beat PCA and variance where the discriminative signal is not the loudest, and remove up to 95% of the bands on biomedical imaging without loss. A simple label-using check flags, in advance, the strongly redundant datasets on which every subset performs alike. Future work will tighten the bottleneck and study attribution less sensitive to channel variance.

ACKNOWLEDGMENT

Financial support from the European Union NAS SAR-101087403 and the United States R44 HL120511 awards is gratefully acknowledged. Narek Chilingaryan and Kristina Ghahramanyan are thanked for providing labelled datasets that included lichen samples and collagen sponges.

REFERENCES

[1] M. F. Balın, A. Abid, and J. Zou, "Concrete autoencoders: Differentiable feature selection and reconstruction," in *Proc. ICML*, 2019.
[2] H. Peng, F. Long, and C. Ding, "Feature selection based on mutual information," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 8, pp. 1226–1238, 2005.
[3] G. Brown, A. Pocock, M.-J. Zhao, and M. Luján, "Conditional likelihood maximisation: A unifying framework for information theoretic feature selection," *J. Mach. Learn. Res.*, vol. 13, pp. 27–66, 2012.
[4] X. He, D. Cai, and P. Niyogi, "Laplacian score for feature selection," in *Proc. NeurIPS*, 2005.
[5] Y. Cai, X. Liu, and Z. Cai, "BS-Nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, 2020.
[6] T. N. Alotaiby, F. E. Abd El-Samie, S. A. Alshebeili, and I. Ahmad, "A review of channel selection algorithms for EEG signal processing," *EURASIP J. Adv. Signal Process.*, vol. 2015, art. 66, 2015.
[7] F. M. Shiri, T. Perumal, N. Mustapha, and R. Mohamed, "Deep learning and federated learning in human activity recognition with sensor data: A comprehensive review," *Comput. Model. Eng. Sci.*, vol. 145, no. 2, pp. 1389–1478, 2025.
[8] M. Rajabinasab, M. E. Houle, O. Chelly, and A. Zimek, "Worse than random? The importance of a baseline for unsupervised feature selection," *arXiv:2605.22973*, 2026.
[9] M. Tangermann *et al.*, "Review of the BCI Competition IV," *Front. Neurosci.*, vol. 6, art. 55, 2012.